CA/Browser Forum Tracker

Part 5 - Revocation and Transparency

Last updated: 2026-11-01

About This Report

This is an experimental, AI-driven research project from SRL. SRL is independent, with no outside funding and no sponsors. This tracker is a summarized pull of the public record, kept as a coherent, concise news update. It draws on the CA/Browser Forum's own ballots, mailing lists and published documents, and on the public sources of the browser root programs. An AI model reads those sources and writes the summary, and every statement links to the source it came from. Where two sources disagree, both are reported. AI makes mistakes, so check the linked source before relying on any statement. The tracker is updated each week that the public record shows a change.

Revocation and transparency are the two ways the web recovers from a certificate that should not exist. Revocation withdraws trust from a certificate already issued. Transparency makes every issued certificate public, so that a bad one can be found at all.

Both were tested hard in the nine weeks to 1 November, and hardest in the last two. Section 1 covers revocation, where a key compromise at eMudhra and GoDaddy's mass revocation of more than 400,000 certificates were followed by a new cluster of bugs. The new bugs concern authorities blocking the tools that fetch their revocation lists. Section 2 covers Certificate Transparency, where Chrome changed its policy on 1 November and an argument over how fresh a log's checkpoint must be is still open. Section 3 covers Merkle Tree Certificates, a design that folds transparency into issuance itself.

1. Revocation Mechanisms and Timelines

Revocation is the part of the certificate system that has never worked well. The industry's answer has been to route around it with shorter certificates, while the deadlines for revoking a bad certificate stay strict.

Recent Developments

The sharpest test came from a key compromise. On 17 October the private keys of eight eMudhra intermediate authorities were found to be recoverable, because their ECDSA private scalars were smaller than 256 [1]. Browser-side blocking did the first work. Chrome blocked the keys through its revocation list within two hours of the report, and Mozilla's block was fully distributed within three [2] [3]. eMudhra revoked the eight certificates the next morning for key compromise, and Sectigo revoked its cross-certificates for them on 19 October [1].

The compromise exposed the limits of the authority's own revocation. eMudhra admitted that its revocation lists had been re-signed with the flawed keys, so an attacker holding the keys could have issued a clean list [1]. A second list, for emSign Root TLS CA - G3, was also affected. The reporter said on 1 November that the preliminary report on the lists, due 29 October, had not been filed [1].

Mozilla removed the root above the eight on 20 October, and Part 4 records the other root program responses [1]. Let’s Encrypt’s issuance software gained a check for small ECDSA scalars on 20 October, the first authority-side code response [4].

The largest event by count ended the week after. On 15 October GoDaddy found that 404,607 certificates from its GoDaddy and Starfield hierarchies carried policy information that did not match its own practice statement [5]. The Baseline Requirements gave it five days, to 18 October at 23:45 UTC, and it said on 16 October that it would not make that deadline [6]. On 22 October it reported the work complete, with a corrected total of 403,643 certificates, 378,067 revoked and 25,576 expired, leaving none valid [6]. Its full incident report followed on 29 October and confirmed those counts [5].

A new cluster of bugs this week concerns authorities blocking the tools that fetch their revocation data.

eMudhra’s firewall rule, in place since September 2022 and removed on 26 October, blocked curl, Wget and similar tools on its list, status and repository hosts [7]. Izenpe fixed the same fault on its status host on 26 October [8]. Microsoft found that the revocation list for its Azure TLS issuing authority on www.microsoft.com refused Wget, under a protection rule in place for over two years. Thirty authorities publish lists on that host [9]. Microsoft targets an exception for 3 November [9].

Sectigo’s case is disputed. Its China list hosts, 180 list addresses in all, refused some tools because a content network’s browser check was on, until a fix on 27 October [10]. Sectigo says user-agent blocking does not breach the Baseline Requirements wording on publishing lists, though it agrees it should not happen, and it wants Forum discussion [10].

Other list and status bugs kept arriving. HARICA published two truncated lists after a disk filled on 29 October, with a full report due 11 November [8]. KIR issued lists whose next-update time was exactly twelve months after the issue time, and Let’s Encrypt argued that this is a point in time, not a validity period [11]. certSIGN reported lists one second or slightly more over the twelve-month limit [11]. Atos and FNMT reported status responder certificate profile faults, which joins seven earlier bugs on the same theme [11] [12].

DigiCert’s earlier report of a revocation list whose sequence number went backwards now has a full report, which traces it to two partitioned lists for one Cisco Meraki authority [13]. DigiCert’s commitment to a twenty-hour regeneration schedule sat in a comment it says went to the wrong incident, so that commitment is unconfirmed [13]. Telekom’s DFN and SwissSign had earlier reported lists whose validity ran one second long, a defect the common linters accept [12]. SHECA reported status responder keys with a public exponent of three, which HARICA called practically unrevocable, and revoked every authority under four roots that browsers do not trust [14].

Revocation work also finished elsewhere. ACCV revoked its dual-purpose subordinate certificate on 29 October, two days ahead of its promise [15]. eMudhra revoked 8,288 unexpired certificates from 19 October, with the last batch about two minutes late [16].

Missed deadlines were already a theme in September. The SHA-1 sunset required unexpired subordinate certificates with SHA-1 signatures to be revoked by 15 September [17]. Within three

days, GoDaddy, Firmaprofesional, FNMT and DigiCert each had a delayed revocation report open [17] [18]. October then brought thirteen bugs about faulty online status responses between 8 and 13 October [19].

The country-code registry hijacks of early October showed browser-side revocation at work. Chrome blocked the unauthorized certificates for Google properties through CRLSets at once, then worked with the issuers to revoke them for other clients [20]. A community count found 102 certificates from the hijacks, all present in a CRLSet of 10,823 entries [21]. Chrome warned that browser-side blocking may not cover every domain [20].

Let’s Encrypt moved its own lifetimes ahead of the Forum’s schedule. On 7 October it announced that its default certificates will last 64 days from 10 February 2027, down from 90 [22]. Subscribers may choose 45 or six days instead, and no valid certificate will be revoked in the change [22]. Staging switched to 64-day certificates on 14 October [23]. The 10 February date was not re-verified this week, because the announcement page returned an error, and no newer schedule has appeared [24].

Let’s Encrypt is also adding validation vantage points. Its staging environment has used six remote network perspectives since 27 October, and production follows on Tuesday 10 November, assuming staging goes well [25]. The Baseline Requirements require at least five remote perspectives from 15 December 2026 [25].

Current State

The online status protocol is being retired rather than repaired. Let’s Encrypt ended its responder service in August 2025, citing privacy, because the protocol tells the responder which site a user is visiting [26]. HARICA stopped putting responder addresses in new web certificates in March 2026 and expects to shut down its web responders by 4 May 2027 [27]. The IETF published a lightweight status protocol profile for high-volume environments, RFC 9919, in July 2026 [28].

Short lifetimes are the replacement. Let’s Encrypt issued its first six-day certificate in February 2025, and those certificates carry no revocation addresses at all [29]. Renewal is steered by the ACME renewal information protocol, published as RFC 9773 in June 2025, which lets an authority tell clients when to renew [30]. Part 1 covers the Forum’s own lifetime schedule.

Browser revocation summaries differ widely in coverage. Firefox’s CRLite has covered 99.84% of certificates visible in transparency logs since August 2025, with about 300 kB of daily updates checked locally [31]. Chrome’s CRLSets carry about 1% of all revocations, aimed at high-impact and emergency cases [31]. The eMudhra response used both, and the compromised keys were blocked in every current Firefox and Chrome before the authority had revoked anything [2] [3]. No CRLite bug changed this week, though that rests on a single search [32].

Mass revocation now has a written template. Mozilla merged an illustrative plan template on 23 September for the mass revocation plans its policy requires [33]. It defines a mass revocation event as at least 100 certificates or 1% of an authority’s active web certificates [34]. Plans must be tested at least every 12 months [34]. GoDaddy’s 403,643 certificates and HARICA’s 63,525 in July 2026 are the two largest such events of the year [6] [35].

The meaning of revocation itself is under discussion. SC-107 would count a certificate as revoked once its serial number appears in a revocation list or draws a revoked status response, from 15 February 2027 [36]. It would also set how quickly an authority must judge a problem report [36]. The ballot was still in discussion on 25 October, past its scheduled end, and Part 1 follows it [36].

Unresolved Conflicts

Three questions sit open. The first is whether blocking tools by name on list and status hosts breaches the Baseline Requirements. Sectigo says its wording on publishing does not forbid it, while Izenpe and Microsoft fixed or planned to fix it [10] [8]. A ruling from a root program, or a Forum ballot that names the practice, would settle it.

The second is when Mozilla's mass revocation testing became auditable. One Mozilla policy change describes annual tests that revoke 30 random certificates within five days, with audit attestation after 15 April 2026 [37]. Mozilla's template and its earlier guidance date third-party assessment from the audit cycle beginning June 2025 [34] [38]. The current text of the policy's revocation section would settle it.

The third is whether the Forum's revocation deadlines will split. Working group discussions in 2025 proposed separate tracks for breaches of the Baseline Requirements and for deviations from an authority's own policy, with 90 days for the latter [39]. GoDaddy's mismatch with its own practice statement ran on the five-day clock under the current rules, and took nine [5] [6]. A ballot on the split entering discussion would settle the question.

2. Certificate Transparency

Every publicly trusted web certificate must appear in public logs, and the browsers admit, monitor and retire the logs themselves. The system is midway through a move from its original design to a static, tiled one that can be served from ordinary storage.

Recent Developments

Chrome changed its policy on 1 November. Announced on 30 October, the change lets logs of the original design refuse final certificates at one submission point while still accepting precertificates, to cut load during the migration [40]. A checkpoint older than 60 seconds now counts as a failed request in Chrome's availability figures, and it becomes an incident only if older than an hour [40]. From 1 January 2027 Chrome will accept no new logs of the original design, including replacement shards [40].

The eMudhra compromise turned the logs into a target. Once the keys were public, people used them to mint certificates and submit them to logs as proofs of concept [41]. By 25 October 107,107 such certificates had been logged on 20 and 21 October, 106,050 of them carrying a Google name [41]. On 20 October Cloudflare closed the submission interfaces of its Raio logs of the original design "to mitigate the eMudhra DoS vector" and asked Apple to reject the three affected shards [42]. Chrome had already moved them from pending to rejected, and nothing newer has appeared [42].

Let’s Encrypt reported a policy breach of its own. Three of its logs, Twig, Sycamore and Willow, served stale checkpoints during six instance replacements of about 2.5 minutes each on 19 and 20 October [43]. One of those replacements deployed the Sunlight log software version that blocks the eMudhra keys [43]. The Chrome transparency team said on 21 October that Chrome policy had been violated [43]. Let’s Encrypt announced a further short outage of its logs for kernel updates on 29 October [44].

Chrome’s log list reached version 95.19 on 1 November, holding 84 logs from 11 operators [45]. Of these, 43 are usable, 36 qualified, 3 retired and 2 read-only [45]. No log has changed state since 14 October, and the list’s earlier movement was all growth [45]. GoDaddy became a log operator when its five Aquamarine tiled logs qualified on 30 September, and sixteen more logs qualified on 13 and 14 October, from HARICA, Sectigo and Let’s Encrypt [45]. Apple’s list rejected fifteen shards covering the first half of 2026 on 5 October and was not re-read this week [46].

A new operator is arriving. On 30 October SwissSign launched two logs on the Tessera software, Eiger as its production log and Rotstock as a test log [47]. SwissSign says it plans to apply to the Chrome and Apple programs, and it is not yet on Chrome’s list [47] [45].

TrustAsia answered the complaints about its logs. On 18 October it hard-coded its block size [48]. On 24 October it said it will launch no logs of the original design after its 2027 shards, keeping the current ones on security updates only [48]. The request that its logs say when a rejected client may retry was never addressed [48].

The most serious operational incident is still HARICA’s. Its new Selene logs missed their maximum merge delay on 12 October and again on 14 October [49]. Its preliminary report of 15 October does not identify the crash cause, because the server logs were missing [49].

A new incident began on 27 October at about 19:15 UTC, with error responses and failures to upload checkpoints, during a traffic surge [50]. HARICA applied per-address rate limits and said service stabilized by the next morning [50]. Earlier in the autumn Geomys, IPng Networks, GoDaddy and Microsec each reported shorter outages, and IPng’s static logs were unreachable again for about 12 minutes on 27 October [51] [52] [53] [54] [55].

Current State

Both log designs are accepted, and neither is retired, though the original design is now closed to new logs from 2027. Chrome admits logs built to the original RFC 6962 design or to the static specification [56] [57]. Since 15 April 2026 a certificate may carry timestamps from static logs alone [58]. Chrome will also stop accepting timestamps delivered through status responses in Chrome 148 [59].

The static logs carry a new obligation. From 1 January 2027 they must let a well-behaved client on a single address read entries at least 1.5 times the log’s growth rate [40]. Existing logs of the original design that Chrome already recognizes are unaffected [40].

Apple states its terms for both designs. Its log program page, published 3 September 2026, accepts logs built to version 1.0.0 of the static specification as well as the original design [60]. Both kinds must meet 99% uptime [60]. An original-design log may take up to 24 hours to merge

a certificate, while a static log may take at most one minute [60]. Apple's repository has no commit after 16 October and no revised policy, and a question of 19 October on when one will appear has drawn no reply [61] [62].

Obligations on both sides are tightening. Chrome's root program has required authorities to log every web server precertificate since 15 June 2026 [63]. From 1 November, every newly added log must publish a machine-readable record of its operational state, referenced from its operator's record [56]. Chrome's policy runs each log through the states pending, qualified, usable, read-only, retired and rejected [64].

Unresolved Conflicts

The sharpest dispute is how fresh a static log's checkpoint must be. Let's Encrypt argued that Apple's and Chrome's one-minute merge delay, read with the original log specification, means a static log can never be down for more than 60 consecutive seconds. That would override the 99% uptime figure, and Let's Encrypt asked whether that is intended [43]. The software's author replied that one-minute merge delays "are simply a mistake" and proposed a 24-hour limit, with a rule that each checkpoint must include everything accepted before it [43].

Others disagree with the author. Andrew Ayer opposed a 24-hour limit because it would allow delays like those of a content network, and proposed treating any checkpoint older than the limit as downtime [43]. Chrome's 30 October change is the only policy response so far, and it treats a 60-second-old checkpoint as a failed request without calling it an incident [40]. Apple has not replied [43]. A joint statement from Apple and Chrome on the merge delay would settle it.

One operational question is also open. HARICA's Selene logs remain qualified after their merge delay misses and a second incident, and Chrome has posted no decision on them [45] [49] [50].

3. Merkle Tree Certificates

A Merkle Tree Certificate replaces the usual chain of signatures with a proof that the certificate appears in a signed public log. Transparency stops being a check after issuance and becomes part of issuance itself.

Recent Developments

The design's specifications moved quickly in the autumn. Version 06 of the IETF draft appeared on 21 September, with authors from Google, Apple, Cloudflare and Geomys [65]. Version 07 followed on 7 October, and no newer revision had appeared by 1 November [66]. The same day, the C2SP project tagged the family of log specifications the design rests on, with the Merkle certificate log specification at version 0.1.0 [67]. That family covers tiles, mirrors, cosignatures, checkpoints and witnesses [67].

Chrome has now published a new draft of its quantum-resistant root program, version 0.4.0, last updated 28 October and aligned with revision 07 [68]. It specifies three to six ML-DSA-44 cosigner keys per authority, subscriber certificates valid for 7 or 47 days, and reuse of domain validation for no more than 10 days [68]. It exempts authorities issuing these certificates from the annual Baseline Requirements audits, though Chrome may ask for operational artifacts

instead [68]. Before that, on 16 October, Chrome pinned its testing program to revision 07, so new cosigner requests must follow it [69].

Apple's own draft policy for these certificates has slipped. Apple wrote on 30 October that a WebPKI incident shifted the team's focus. The draft will not appear by the end of October and will come as soon as possible, with no new date [70].

Issuers are building to it. As of 25 October, Let's Encrypt's issuance software was landing code for Merkle certificate trees, including reading checkpoints from tiles and the use of relative object identifiers [4]. Cloudflare says its planned authority will run mirroring cosigners and a new family of static transparency logs [71]. Chrome's testing program already requires every participant to operate both a log and a mirror [72].

Current State

None of this is yet a production requirement. Chrome plans the production phase of its quantum-resistant program for 2027 [72]. Part 4 covers the root program policies that will decide which of these authorities browsers trust.

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